

Construction Project Management System (CPMS)

Official MERN Stack Development Plan – Evaluator Documentation

SUBMITTED TO

Project Evaluation Committee

PROGRAM TRACK

ITI ICC – MERN Stack (MongoDB, Express, React, Node)

PROJECT ROADMAP

May 25, 2026 - June 16, 2026 (16 Working Days)

1. Project Scope & Tech Stack Strategy

This plan details the Agile execution blueprint for the **Construction Project Management System (CPMS)**. To keep development simple, practical, and highly effective for graduation review, the architecture has been streamlined around standard MERN stack patterns:

- **Strict Epic Sequencing:** Software implementation initiates on May 31st and tightly respects the execution flow: **Epic 2 (Upload & AI) → Epic 3 (Contract Data Review) → Epic 1 (Auth Infra) → Epic 4 (Planned Budget) → Epic 5 (Actual Progress) → Epic 6 (Reports & KPIs)**.

2. High-Level Agile Release Matrix

Phase	Date Window	Working Days	Core Delivery Target (Milestones)
Sprint 1	May 25 - May 30	5 Days	No Code Implementation. Complete UI/UX Figma wireframes, functional SRS test mapping, MongoDB Mongoose schema designs, and API routing specs.
Sprint 2	May 31 - June 06	5 Days	Implementation Phase A (Epic 2 & 3): Build Express file upload routes, S3 document storage helper, direct LLM API calling controllers, and contract dashboard interfaces.
Sprint 3	June 09 - June 13	5 Days	Implementation Phase B (Epic 1 & 4): Set up MongoDB collections, secure JWT token routes, RBAC middleware logic, and the milestone budget auto-allocation service.
Sprint 4	June 15 - --June 16	2 Days	Implementation Phase C (Epic 5 & 6): Site progress data submission forms, PMO authorization rules, MongoDB aggregations for KPIs (SPI/CPI), and spreadsheet exports.

3. Comprehensive Sprint Task Management

Sprint 1: Architecture Blueprinting & UI/UX Design (May 25 – May 30)

Sprint Goal: Complete all visual assets, MongoDB document collections blueprints, and testing schemas. Absolutely no code implementation is performed by any team member during this phase.

Task ID	Task Name & Functional Scope	Role	Assignee	Priority
CPMS-101	MongoDB Document Collection & Mongoose Schema Mapping Blueprint	BACKEND	D2 (BE-Only)	HIGH
CPMS-102	Cloud Storage Buckets & Presigned URL Integration Design	BACKEND	D2 (BE-Only)	HIGH
CPMS-103	LLM Prompt Constraints Matrix & Direct AI JSON Response Spec	FULLSTACK	D3 (Balanced)	HIGH
CPMS-104	Milestone Budget Allocation Algorithm Technical Specification	FULLSTACK	D4 (Balanced)	HIGH
CPMS-105	Client-Side Component Tree Layout & Routing Documentation	FRONTEND	D1 (FE-Only)	HIGH
CPMS-106	Design System System Selection (Tailwind Configuration Layouts)	FRONTEND	D5 (Lean-FE)	HIGH
UX-101	Figma High-Fidelity Design Specifications (All Core Views)	DESIGN	UX Designer	HIGH
QA-101	Operational Test Suite Matrices Design (Module 1, 2, 3 Cases)	TESTING	T1 & T2 (QA)	HIGH

Sprint 2: Code Implementation Kickoff — Core Contract AI Pipeline (May 31 – June 06)

Sprint Goal: Initiate software engineering. Deliver standard direct contract upload, secure file storage, and integrated server-side AI parsing methods alongside data tables.

Task ID	Task Name & Functional Scope	Role	Assignee	Priority
CPMS-201	Epic 2: POST /api/contracts/upload File Handler & S3 Ingest Logic	BACKEND	D2 (BE-Only)	HIGH
CPMS-202	Epic 2: Server-Side Contract Text Extractor Utility Implementation	BACKEND	D2 (BE-Only)	HIGH
CPMS-203	Epic 2: Direct Server-to-LLM Controller (Call, Receive, Validate 13 JSON fields)	FULLSTACK	D3 (Balanced)	HIGH
CPMS-204	Epic 3: GET /api/contracts Database Query Controller with Filters	FULLSTACK	D4 (Balanced)	MED
CPMS-205	Epic 3: GET /api/contracts/:id Detail Node & Milestone Sorters	FULLSTACK	D4 (Balanced)	MED
CPMS-206	Epic 2: React Project Scaffold, File Uploader & Async Processing UI View	FRONTEND	D1 (FE-Only)	HIGH
CPMS-207	Epic 3: Core Contract Registry Data-Table & Filter Dashboard Layouts	FRONTEND	D5 (Lean-FE)	HIGH
UX-201	Figma Interactive Forms, Approval Grids & Performance Report Design	DESIGN	UX Designer	HIGH
QA-201	Mongoose Schema Sanity Setup & Integration Testing Environment Setup	TESTING	T1 & T2 (QA)	HIGH

Sprint 3: User Security & Automated Budget Partitioning (June 09 – June 13)

Sprint Goal: Deploy application authentication and user roles. Implement the dynamic Mongoose document sub-generators that allocate funds into project schedules.

Task ID	Task Name & Functional Scope	Role	Assignee	Priority
CPMS-301	Epic 1: User Collection Mongoose Indexes & Salt-Hashing Logic	BACKEND	D2 (BE-Only)	HIGH
CPMS-302	Epic 1: POST /api/auth/login Endpoint (JWT Token & Cookies Token Delivery)	FULLSTACK	D3 (Balanced)	HIGH
CPMS-303	Epic 1: Server-Side RBAC Route Interceptor Middleware Logic	FULLSTACK	D3 (Balanced)	HIGH
CPMS-304	Epic 4: Contract Mongoose Document Budget Allocation Auto-Split Service	FULLSTACK	D4 (Balanced)	HIGH
CPMS-305	Epic 6 (Risk Mitigation): MongoDB Aggregation Pipeline Scripts Pre-Compilation	FULLSTACK	D4 (Balanced)	HIGH
CPMS-306	Epic 3: React 13-Field Extracted Contract Fields Review & Auditing Form	FRONTEND	D1 (FE-Only)	HIGH
CPMS-307	Epic 4: Planned Target Matrix & Dynamic Variance Table Dashboard Elements	FRONTEND	D5 (Lean-FE)	HIGH
QA-301	Functional Direct Upload Verification Cycles (AI Response Matrix Checking)	TESTING	T1 & T2 (QA)	HIGH

Sprint 4: Operational Tracking & Analytical KPI Evaluation (June 15 – June 16)

Sprint Goal: Deploy project field progress logging views. Wire optimized MongoDB aggregate report utilities and finalize comprehensive system presentation passes.

Task ID	Task Name & Functional Scope	Role	Assignee	Priority
CPMS-401	Epic 5: Progress Entry Post Route & PMO Status Approvals Actions Controller	BACKEND	D2 (BE-Only)	HIGH
CPMS-402	Epic 6: Analytical Performance Reports Router via Pre-Built Pipelines	FULLSTACK	D4 (Balanced)	HIGH
CPMS-403	Epic 6: Excel Data Sheet Workbook Streamer Controller	FULLSTACK	D3 (Balanced)	MED
CPMS-404	Epic 6: Main Dashboard Panel Live Core Mongoose Aggregate Data Binding	FULLSTACK	D3 (Balanced)	MED
CPMS-405	Epic 5: PMO Operations Log Pending Review & Approvals Interface Screen	FRONTEND	D1 (FE-Only)	HIGH
CPMS-406	Epic 6: Consolidated Performance Report Sheets View Components	FRONTEND	D5 (Lean-FE)	HIGH
QA-401	Full End-to-End MERN Stack System Evaluation Walkthrough Sanity Pass	TESTING	T1 & T2 (QA)	HIGH

4. Resource Matrix & Operational Ownership

D1 — Frontend Software Engineer (FE-Only)

Role: Manage complex user interfaces, routing contexts, input fields validations, and client application workflows.

Tasks: CPMS-105 (Routing Layout Design) → CPMS-206 (File Uploader Views) → CPMS-306 (13-Field Verification Form) → CPMS-405 (PMO Operations Sheet).

D2 — Backend Software Engineer (BE-Only)

Role: Lead MongoDB Mongoose collection modeling, indexing optimizations, file buffers streams, and status change endpoints.

Tasks: CPMS-101 (Mongoose Collection Topology Specifications), CPMS-102 (S3 Buckets Integrations Spec) → CPMS-201 (S3 Upload API Node), CPMS-202 (Text Extraction Utility Controller) → CPMS-301 (Mongoose Database Indexing Rules) → CPMS-401 (PMO Approvals API Node).

D3 — Full-Stack Software Engineer (Balanced)

Role: Own primary server AI handlers controllers, native JWT session management interceptors, and data stream conversions.

Tasks: CPMS-103 (Direct LLM Integration JSON Contract Specifications) → CPMS-203 (Direct LLM Controller Method Implementation) → CPMS-302 (JWT Authentication Node), CPMS-303 (RBAC Access Route Middlewares) → CPMS-403 (Excel Workbook Data Exporter), CPMS-404 (Main Dashboard Mongoose Aggregates Binding).

D4 — Full-Stack Software Engineer (Balanced)

Role: Lead complex MongoDB aggregate analytics scripting pipeline structures and chronological arithmetic modules.

Tasks: CPMS-104 (Milestone Split Metrics Logic Matrix Architecture Specs) → CPMS-204 (Contract Filtering Query Controller), CPMS-205 (Chronological Data Array Sorting Router) → CPMS-304 (Auto-Distribution Service Engine), CPMS-305 (Risk-Mitigation MongoDB Aggregate Pre-Compile Sheets) → CPMS-402 (Analytical Performance Aggregate Routers).

D5 — Frontend Software Engineer (Leans FE)

Role: Maintain clean layout design tokens systems, global styling wrappers rules, tabular charts layouts, and validation badges indicators.

Tasks: CPMS-106 (Design Tokens Definitions Specs) → CPMS-207 (Core Registry Screen) → CPMS-307 (Baseline Dynamic Budget Variance Comparison Grids) → CPMS-406 (Analytical Status Summary Sheets View).